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CLINICAL DIABETOLOGY

**ISSN:** 2450-7458

**e-ISSN:** 2450-8187

# Short-Term Blood Glucose Reduction After Bodyweight Soleus Push-Up in Type 2 Diabetes

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**DOI:** 10.5603/cd.114889

**Article type:** Observation letter

**Submitted:** 2026-09-19

**Accepted:** 2026-10-05

**Published online:** 2026-10-08

This article has been peer reviewed and published immediately upon acceptance.
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OBSERVATION LETTER

Short title: Bodyweight Soleus Push-Up in T2D

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DOI: 10.5603/cd.114889

Received: 19.09.2026 Accepted: 5.10.2026 Published: 8.10.2026

## Introduction

Managing type 2 diabetes (T2D) requires feasible activity strategies that can be used by patients who have difficulty sustaining conventional exercise. Current recommendations support physical activity and reduced sedentary time as part of T2D care [1]. Seated soleus-dominant activity has recently attracted attention because it may increase local oxidative metabolism and improve glucose regulation while requiring little space or equipment [2].

## Methods

We conducted a single-center, nonrandomized quasi-experimental observation at a community health center in Kendari, Indonesia. Thirty-six adults with clinically diagnosed T2D were allocated to a bodyweight soleus push-up group (n = 18) or a non-weighted soleus push-up group (n = 18). Participants performed the assigned movement for 15 minutes per session, three times weekly, for three consecutive weeks. Capillary blood glucose was measured before and after the intervention period. Sessions were conducted in the morning before antidiabetic medication intake to reduce timing-related variation. The study was approved by an institutional health research ethics committee (No. 199/KEP-UMW/X/2025), and all participants provided written informed consent.

### Statistical analysis

Continuous data were summarized as the mean ± standard deviation. Within-group changes were analyzed using paired t-tests, and between-group reductions were compared using an independent-samples t-test. We used 95% confidence intervals (CIs) and Cohen's d to describe estimate precision and effect size.

## Results

Baseline characteristics were broadly comparable: mean age was 54.2 ± 5.79 years in the bodyweight group and 55.1 ± 4.40 years in the non-weighted group; both groups included 2 men and 16 women; and diabetes duration was similar (6.33 ± 2.43 vs. 6.28 ± 2.40 years). Both groups had lower capillary blood glucose after three weeks. The reduction was larger in the bodyweight soleus push-up group than in the non-weighted group (74.5 ± 14.2 mg/dL vs. 43.9 ± 12.7 mg/dL; between-group mean difference, 30.6 mg/dL; 95% CI: 21.5–39.7; $p < 0.001$; Cohen's d = 2.36; Tab. 1).

## Discussion

This preliminary observation suggests that adding bodyweight loading to a seated soleus push-up may produce a greater short-term glucose-lowering response than the same movement without additional loading. The finding is biologically plausible because skeletal muscle contraction can

increase glucose uptake through insulin-independent pathways [3], and brief light activity or simple resistance breaks can improve glycemic responses in adults with T2D [4]. A recent soleus push-up study in patients with T2D also reported favorable effects on blood glucose [5]. However, these findings should be interpreted cautiously. The study was small, single-center, and nonrandomized; diet, medication use, and habitual activity were not fully controlled; follow-up was short; and glycated hemoglobin (HbA1c) was not assessed. This low-cost seated exercise warrants evaluation in adequately powered, registered randomized studies with longer follow-up and clinically relevant outcomes.

## Article information

### Data availability statement

The data are available from the corresponding author upon reasonable request.

### Ethics statement

The study was approved by an institutional health research ethics committee (No. 199/KEP-UMW/X/2025), and all participants provided written informed consent.

### Author contributions

C.H. Rosjidi: conception/design; drafting; critical revision, final approval; Narmawan: conception/design: critical revision, study materials/patients; final approval; Risnawati: drafting; study materials/patients; final approval; S. Umrana: critical revision; study materials/patients; final approval; M. Syahwal: data analysis/interpretation; statistical expertise; final approval; Narmi: drafting: study materials/patients; final approval; Mien: critical revision; final approval; Hasrima: study materials/patients; final approval; Herman: data analysis/interpretation, statistical expertise, critical revision; final approval; Mubarak: conception/design; critical revision; final approval; Rahmawati A.: drafting: study materials/patients; final approval

### Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

## Acknowledgments

The authors would like to thank the management and staff of the community health centre where the study was conducted for their assistance during data collection. The authors also express their appreciation to all participants for their time, cooperation, and contribution to this study.

## Conflict of interest

The authors declare no conflict of interest.

## Supplementary material

No supplementary material was submitted with this manuscript.

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**Table 1.** Change in capillary blood glucose after three weeks of intervention

| Group | Before intervention [mg/dL] | After intervention [mg/dL] | Mean reduction; mean ± SD (95% CI) [mg/dL] |
|---|---|---|---|
| Non-weighted soleus push-up | 248.6 ± 22.1 | 204.7 ± 18.9 | 43.9 ± 12.7 (37.6–50.2) |
| Bodyweight soleus push-up | 251.2 ± 23.4 | 176.7 ± 19.3 | 74.5 ± 14.2 (67.4–81.6) |

Values are presented as mean ± SD unless otherwise stated. The between-group mean difference in reduction was 30.6 mg/dL (95% CI: 21.5–39.7; $p < 0.001$)

CI — confidence interval; SD — standard deviation